Operational Negatives — Preregistration Package

Subtitle: Licensed inverses, torsor specificity, and closure diagnostics for negative-number reasoning
Project stack: hybrid experimental design + GLMMs + Bayesian two-policy mixture
Artifacts: generator.py • two_policy_mixture.stan • absence_frame_pilot.md • sample_trials_frame_A_seed2026.json

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0) Executive summary

Thesis. Negatives are *licensed inverses* of an operational practice. Participants perform best when the frame makes the inverse role computable (netting to a neutral). They perform worst when “minus” is framed as absence/removal (partial, non-invertible). When only differences are meaningful (torsor), absolute queries suffer while difference queries remain strong. A closure diagnostic (parenthesization sensitivity) captures whether a frame induces an order-sensitive, non-closed internal operation.

Design overview (hybrid). Between-subjects for **Frame** (A Netting-Debt, B Netting-Displacement, C Pure-Symbol minimally licensed control, D Absence/Removal). Within-subjects for **Origin** (shown/hidden, i.e., torsor toggle), **Parenthesization** (associator), and an **Inventory Completion** mini-block (no-backorders vs backorders). Trial budget \~40–45 per participant (\~18–22 min).

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1) Theoretical commitments (clarified)

- **Licensing criterion.** A frame licenses negatives iff the participant’s implemented operation is or is treated as **group-like**; otherwise “−” degenerates to **removal** in a partial algebra.
- **Torsor vs group.** In a torsor, inversion lives in the **acting group**; there is no distinguished origin, only differences. Hiding the origin should selectively impair **absolute** queries in netting frames while leaving **difference** queries intact.
- **Completion.** When a practice demands netting across gains/losses (e.g., backorders), the monoid **completes** to an abelian group (deficits become stable states).
- **Closure diagnostic.** If a frame induces order sensitivity ($(x \oplus y) \oplus z \ / \ x \oplus (y \oplus z)$ in behavior), it reveals missing state/closure; netting frames should minimize this.

Frame C (Pure-Symbol) — baseline status. C is the **minimally licensed control**: standard rules allow inverses but the UI provides no netting affordance or neutral substrate. Prediction: C > D but C < A/B.

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2) Hypotheses (final text)

- **H1a (Netting vs Symbol):** (A,B) > C on accuracy; RT (A,B) < C.
- **H1b (Symbol vs Absence):** C > D on accuracy; RT C < D.
- **H2 (Torsor specificity; preregistered three-way interaction). Frame × Origin (shown/hidden) × Query-Type (absolute/difference)**.

- In **A/B**, hiding the origin selectively impairs **absolute** queries while sparing **difference** queries.
- In **C/D**, this selective impairment is absent or markedly reduced.
- **Key test:** Contrast of selective impairment under OriginHidden for A/B vs C/D.
- **H3 (Completion).** Allowing backorders reduces **clamp-to-zero** errors and improves generalization to novel signed items.
- **H4 (Parenthesization / associator). Simple** Δ_{Assoc} (absolute difference in error rate or mean response deviation between left vs right parenthesization) satisfies: D > C > (A,B). **Distributional** W_1 serves as a secondary, group-level confirmation.

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3) Design & materials

Between-subjects: Frame ∈ {A,B,C,D}. Target $N = 240$ → \~60 per frame. **Primary contrasts:** (A∪B) vs C; C vs D. Pairwise A vs B exploratory.

Within-subjects (inside assigned frame):

- **Origin:** shown vs hidden (torsor toggle), stratified across items.
- **Parenthesization:** 4 matched operand triples → 8 trials (L/R) for the simple associator.
- **Inventory mini-block:** 12 trials (6 no-backorders, 6 backorders).

Trial plan per participant:

- **Main block (28 trials):** 8 signed sums; 10 sign-structure (double negative, subtracting a negative, unary vs binary minus, etc.); 6 comparisons; 4 parenthesization pairs (8 trials included in the 28). Origin and Query-Type are balanced.
- **Inventory block (12 trials):** toggles backorders (completion). “Cannot perform” feedback **only** in **no-backorders**.

Manipulation checks: End-of-block strategy check: two forced-choice (behavioral descriptions) + one free response (30s). Use as a moderator, not exclusion.

Absence/Removal (D) contamination guard: purely verbal/static pictorial prompts; hard floor at zero; no symmetric arrows, no neutral line; correctness feedback only.

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4) Variables & derived metrics

Primary outcomes: accuracy (0/1), RT (ms, log-transformed).
Error taxonomy: sign flip, magnitude error, unary/binary confusion, **clamp-to-zero**.

Clamp-to-zero (operational):

- **Strict:** correct answer < 0 and response == 0.
- **Lenient:** correct answer < 0 and response $\geq$ 0.

Parenthesization effect (simple Δ_{Assoc}). For each participant × frame: absolute difference in mean response deviation (or error rate) between left vs right parenthesization of the same triple. This simple effect (and its Frame interaction) is the primary test; distributional W_1 is secondary and computed at group level via bootstrap.

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5) Analysis plan

Frequentist (primary; fixed-N).

- **Accuracy (GLMM):** `acc ~ frame * origin * query_type + item_type + (1 | participant) + (1 | item)`
  Planned contrasts: H1a (A∪B) > C; H1b C > D; H2 three-way interaction; H4 parenthesization main effect × frame.
- **RT (LMM on log-RT):** same fixed/random structure; report ex-Gaussian sensitivity in appendix if needed.

Bayesian (secondary): two-policy mixture.

- Bernoulli mixture with two logistic channels (**Netting** vs **Absence-like**), mixture weight π varying by Frame and Origin with participant random intercept.
- Priors: $\mathcal{N}(0,1)$ on fixed effects; HalfNormal(0,1) on random-effect SD.
- ROPE for negligible frame differences on logit: ±0.1 (OR $\approx$ [0.90, 1.11]).
- Use WAIC/LOO for model comparison; simulated recovery included.

Stopping. Fixed-N for all frequentist tests. Bayesian fits may be extended **only** for posterior precision (reported separately).

Order & fatigue. Include origin order and inventory order as fixed effects; report interactions. Robustness: drop first 4 trials per block and re-fit.

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6) Falsification map (explicit)

- **Licensing theory fails** if A/B do **not** beat C and C does **not** beat D on accuracy/RT.
- **Torsor specificity fails** if the OriginHidden effect is not selectively larger for **absolute** than **difference** queries in A/B (or if C/D show the same selective pattern).
- **Completion fails** if backorders do **not** reduce clamp-to-zero errors and improve generalization.
- **Closure diagnostic fails** if D $\not>$ C $\not>$ (A,B) on the simple parenthesization effect.

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7) Implementation notes (critical)

- **Frame contrasts:** Primary = (A∪B) vs C; C vs D. A vs B exploratory (power).
- **Uniform feedback:** All main-block trials accept any numeric answer; feedback is ✓/✗ only. “Cannot perform” is restricted to **Inventory/no-backorders**.

- **Strategy check wording (behavioral):**
- “I canceled credits and debts to reach zero.” (Netting)
- “I imagined physically removing things.” (Absence)
- “I followed the math rules I learned in school.” (Pure-Symbol)
- “I only focused on how much things changed.” (Difference)

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8) Artifacts & how to use them

- **Stimulus JSON Generator:** `generator.py`
  Usage: `python generator.py --frame A --seed 2026 --out trials_A_2026.json`
  Outputs per-participant JSON with balanced items, origin toggle, and inventory block. Includes `copy_keys` for frame-specific UI text.

- **Bayesian mixture model scaffold:** `two_policy_mixture.stan`
  Implements the Bernoulli logistic two-channel mixture with frame/origin-dependent mixture weights and participant random intercept. Provides `log_lik` for WAIC/LOO.

- **Absence frame pilot spec:** `absence_frame_pilot.md`
  One-page implementation with 12 pilot items, 2 comprehension checks, and strict contamination guardrails.

- **Sample trials (Frame A):** `sample_trials_frame_A_seed2026.json`
  Demonstration of generator output.

Download links:

- generator.py — sandbox:/mnt/data/generator.py
- two_policy_mixture.stan — sandbox:/mnt/data/two_policy_mixture.stan
- absence_frame_pilot.md — sandbox:/mnt/data/absence_frame_pilot.md
- sample_trials_frame_A_seed2026.json — sandbox:/mnt/data/sample_trials_frame_A_seed2026.json

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9) External references (selected)

- Vlassis, J. (2004). *Making sense of the minus sign or negativity.* (On classic difficulties with “−”, roles, and instruction).
  https://www.sciencedirect.com/science/article/abs/pii/S0959475204000441
- Fischer, M. H., & Shaki, S. (2014). *Do negative numbers have a place on the mental number line?*
  https://www.researchgate.net/publication/228967986_Do_negative_numbers_have_a_place_on_the_mental_number_line
- McNeil, N. M., et al. *Making Sense of Negative Numbers.* GUPEA repository.
  https://gupea.ub.gu.se/handle/2077/24151
- Murray, H. (1991). *Children's Ideas of Negative Numbers.* ERIC.
  https://files.eric.ed.gov/fulltext/ED342632.pdf

- Sneed, G., & colleagues (2024). *Tools, Tricks and Topics Teachers Use for Integer Arithmetic.* EJ RSME. https://ejrsme.icrsme.com/article/view/23771/14961

Notes. These external sources establish (i) the systematic difficulty with negatives, (ii) number-line and spatial encodings, and (iii) the role of context/metaphor. Our contributions add the **practice-first licensing** criterion, **torsor specificity**, and a **closure diagnostic** (behavioral associator) that together make the theory falsifiable.

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10) Ethics & data management

- Adult online participants (18+), informed consent, anonymized data.
- Exclusion: $\geq$2 failed attention checks in any two blocks; trial-level RT trimming (<300 ms or >10 s; 3×IQR within-block).
- Share de-identified data + code + materials post-acceptance.

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11) Timeline

- **Week 1:** Pilot Absence frame; finalize item banks and copy.
- **Week 2:** Launch main study (N=240).
- **Week 3:** Primary analyses (GLMMs), then Bayesian mixture + robustness.

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12) Appendices

A. Parenthesization items (templates)

- L: $(a+b)+c$ vs R: $a+(b+c)$ over matched signed triples (balanced magnitudes / signs); 4 pairs per participant.

B. Negative-number screener (3 items)

1. Place −4 and −7 on a number line; which is larger?
2. What does the minus sign mean in "−x"?
3. Compute $5-(-3)$.

C. Inventory completion examples

- No-backorders: start=5, ship=7 → **cannot perform** (strict), clamp metric stored as 0.
- Backorders: start=5, ship=7 → **−2** (valid state).

Prepared for preregistration and immediate build. This document consolidates the final design, analysis plan, falsification commitments, artifacts, and references.